

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (IT)

December 2011

IT 304 Modeling Simulation & Operational Research

Date: 16.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION - I

Q - 1 (a) Explain utility of Operations Research in engineering. [03]

(b) Briefly explain the characteristics of Operations Research. [04]

Q - 2 (a) Answer the following questions True or False [04]

- (i) Every basic solution represents an extreme point of the convex set of feasible solutions.
- (ii) The slack and surplus variables can be unrestricted in sign.
- (iii) The total number of allocation in a basic feasible solution of transportation problem of  $m \times n$  size is equal to  $m+n-1$
- (iv) If  $k^{\text{th}}$  constraint of the primal LPP is equality, then the  $k^{\text{th}}$  dual variable is unrestricted in sign for dual LPP

(b) Solve the following problem using Two - phase Method. [06]

Maximize  $Z = 4x_1 + x_2$ ; Subject to  $3x_1 + x_2 = 3$ ,  $4x_1 + 3x_2 \geq 6$ ,  $x_1 + 2x_2 \leq 4$ ;  $x_1, x_2 \geq 0$

(c) Solve the following LPP by using graphical method [04]

Maximize  $Z = 4x_1 + 10x_2$ ; Subject to  $2x_1 + x_2 \leq 50$ ,  $2x_1 + 5x_2 \leq 100$ ,  $2x_1 + 3x_2 \leq 90$ ;  
 $x_1 \geq 0, x_2 \geq 0$

## OR

Q - 2 (a) Write the dual of the following LPP: [04]

Maximize  $Z = 14x_1 + 10x_2$

Subject to  $2x_1 + x_2 \leq 6$ ,  $9x_1 + 5x_2 \geq 11$ ,  $2x_1 + 3x_2 \leq 90$ ;  $x_1 \geq 0, x_2 \geq 0$

(b) A firm manufactures two types of products A and B and sells them at a profit of Rs. 2 on type A and Rs. 3 on type B. each product is processed on two machines G and H. type A requires 1 minute of processing time on G and 2 minutes on H; Type B requires 1 minute on G and 1 minute on H. the machine G is available for not more than 6 hours, 40 minutes while H is available for 10 hours during any working day. Formulate this problem as a linear programming problem. [04]

(c) Solve the following LPP by using simplex method [06]

Maximize  $Z = 5x_1 + 3x_2$

Subject to  $x_1 + x_2 \leq 2$ ,  $5x_1 + 2x_2 \leq 10$ ,  $3x_1 + 8x_2 \leq 12$ ;  $x_1 \geq 0, x_2 \geq 0$



- Q - 3 (a)** Describe a method of drawing minimum number of lines in context of assignment problem. Write the name of method. [04]
- (b)** Define (i) pure Integer programming problem (ii) mixed integer programming problem. [02]
- (c)** Determine an initial basic feasible solution to the following transportation problem by using a NWCM method. Also determine optimum solution. [08]

|                | Unit cost (₹) of transportation |                |                |                | Supply |
|----------------|---------------------------------|----------------|----------------|----------------|--------|
|                | D <sub>1</sub>                  | D <sub>2</sub> | D <sub>3</sub> | D <sub>4</sub> |        |
| S <sub>1</sub> | 19                              | 30             | 50             | 10             | 7      |
| S <sub>2</sub> | 70                              | 30             | 40             | 60             | 9      |
| S <sub>3</sub> | 40                              | 8              | 70             | 20             | 18     |

**OR**

- Q - 3 (a)** A company has four sales territories & four salesmen available for assignment. [08]

|            | Territory    |    |     |    |
|------------|--------------|----|-----|----|
|            | Sales '000 ₹ |    |     |    |
|            | I            | II | III | IV |
| Salesman A | 42           | 35 | 20  | 21 |
| B          | 30           | 25 | 20  | 15 |
| C          | 30           | 25 | 20  | 15 |
| D          | 24           | 20 | 16  | 12 |

The sales territories are not equally & rich their sales potential & salesmen also differ in their setting ability. The matrix gives the sales in 'thousand rupees for each salesman to be assigned to each territory.

How should the territories be assigned to salesman, so as to maximize the total sales?

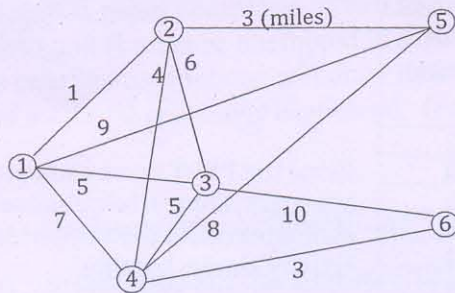
- (b)** Find the Initial Basic Feasible Solution of the Transportation Problem using Vogel's approximation method.

|                | Unit cost (₹) of transportation |                |                |                | Supply |
|----------------|---------------------------------|----------------|----------------|----------------|--------|
|                | D <sub>1</sub>                  | D <sub>2</sub> | D <sub>3</sub> | D <sub>4</sub> |        |
| O <sub>1</sub> | 5                               | 3              | 6              | 2              | 19     |
| O <sub>2</sub> | 4                               | 7              | 9              | 1              | 37     |
| O <sub>3</sub> | 3                               | 4              | 7              | 5              | 34     |
| <b>Demand</b>  | <b>16</b>                       | <b>18</b>      | <b>31</b>      | <b>25</b>      |        |

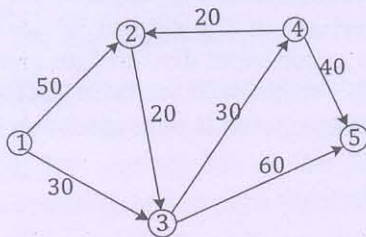
## **SECTION - II**

- Q - 4 (a)** State whether following statements are **True** or **False** [04]
- (i)** A steady state exists in a queue if  $\lambda < \mu$ .
- (ii)** The distribution of arrivals in a queuing system can be considered as a pure birth process.
- (iii)** The designation of Poisson arrival, Exponential service, single server and limited queue selected randomly is represented by (M/M/1): ( $\infty$ /SIFO)
- (iv)** In a simple queuing model the waiting time in the system is given by  $\frac{\mu}{\mu - \lambda}$
- (b)** Explain Kendall's notations for classifying queuing models. [03]

- Q - 5 (a)** Midwest TV Cable Company is in the process of providing cable service to five new housing development areas. Figure given below depicts possible TV linkages among the five areas. The cable miles are shown on each arc. Determine the most economical network. [07]



- (b)** The network in figure below gives permissible routes and their lengths in miles between city 1 and four other cities. Determine the shortest routes between city 1 and each of the remaining four cities. [07]



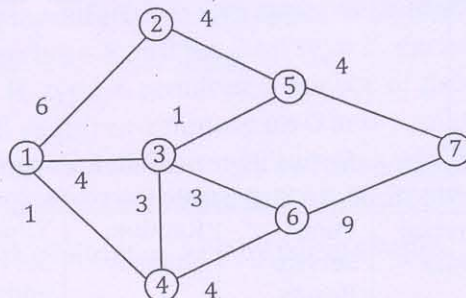
**OR**

- Q - 5 (a)** A small project has 7 activities and the time in days for each activity is given below [07]

| Activity              | A | B | C | D | E   | F        | G |
|-----------------------|---|---|---|---|-----|----------|---|
| Immediate predecessor | - | - | A | A | B,D | B,C or D | E |
| Time in days          | 6 | 8 | 3 | 4 | 6   | 10       | 3 |

Draw the project network and determine the critical path.

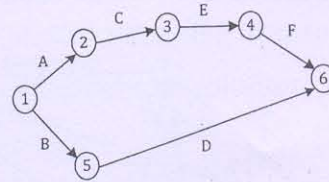
- (b)** For the network given below, find the flow pattern giving the maximum flow from the source to sink, given that the arc capacity from node i to node j is the number nearest node i along the arc between these nodes [07]





- Q - 6 (a)** Sharon Lowe, Vice President for marketing for the Electronic Toys Company, is about to begin a project to design an advertising campaign for a new line of toys. She wants the project completed within 57 days in time to launch the advertising campaign at the beginning of the Christmas season. Sharon has identified the six activities (labeled A, B, ..., F) needed to execute this project. Considering the order in which these activities need to occur, she also has constructed the following project network. [08]

| Activity | Optimistic Estimate (days) | Most Likely Estimate | Pessimistic Estimate (days) |
|----------|----------------------------|----------------------|-----------------------------|
| A        | 12                         | 12                   | 12                          |
| B        | 15                         | 21                   | 39                          |
| C        | 12                         | 15                   | 18                          |
| D        | 18                         | 27                   | 36                          |
| E        | 12                         | 18                   | 24                          |
| F        | 2                          | 5                    | 14                          |



Using the PERT three estimate approach, Sharon has obtained the estimates of the duration of each activity shown in table.

- Find the estimate of the mean and variance of the duration of each activity.
  - Find the mean critical path.
  - Use the mean critical path to find the approximate probability that the advertising campaign will be ready to launch within 57 days.
- (b)** In a railway marshalling yard, goods train arrives at the rate of 30 trains per day. Assume that the inter arrival time follows an exponential distribution and the service time is also to be assumed as exponential with a mean of 36 minutes. Calculate: (i) The probability that the yard is empty (ii) The average queue length, assuming that the line capacity of the yard is 9 trains. [06]

**OR**

- Q - 6 (a)** Automata car wash facility operates with only one bay. Cars arrive according to a Poisson distribution with a mean of 4 cars per hour, and may wait in the facility's parking lot if the bay is busy. The time for washing and cleaning a car is exponential, with a mean of 10 minutes. Cars that cannot park in the lot can wait in the street bordering the wash facility. [06]
- Determine the average number of cars waiting in the queue.
  - Determine the percent utilization of the wash bay.
  - Determine the probability that an arriving car must wait in the parking lot prior to entering the wash bay.

- (b)** The probability distribution of inter arrival time in a single server model is [08]

| x               | 1             | 2             | 3             |
|-----------------|---------------|---------------|---------------|
| $f(x) = P(X=x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

And the distribution of service time is

| y               | 1             | 2             | 3             |
|-----------------|---------------|---------------|---------------|
| $g(y) = P(Y=y)$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{4}$ |

Complete the following table using the two digit random numbers as 12, 40, 48, 93, 61 to generate arrivals and 54, 90, 18, 38, 16 to generate the corresponding service times.

| Arrival Number | Random Number | Arrival Time | Time Service Begins | Random Number | Time Service ends | Waiting time in Queue |
|----------------|---------------|--------------|---------------------|---------------|-------------------|-----------------------|
|----------------|---------------|--------------|---------------------|---------------|-------------------|-----------------------|

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